

First proof: Duality for tropical linear subspaces

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Let M be a matroid, and let $\text{Trop}(M)$ be the corresponding tropical linear space, i.e., the closure of the Bergman fan of M in tropical projective space. The relative Dressian of M is

$$\mathbf{Dr}(M) := \{ \text{tropical linear subspaces of } \text{Trop } M \},$$

which is partially ordered by inclusion of tropical linear spaces.

Suppose M admits an order-reversing involution $F \mapsto F^\perp$ on the lattice of flats $\mathcal{L}(M)$. Note that there is a natural inclusion $\mathcal{L}(M)^{\text{op}} \subset \mathbf{Dr}(M)$ given by $F \mapsto \text{Trop}(M/F \oplus U_{0,F})$.

Question. Is there an order-reversing involution Φ on $\mathbf{Dr}(M)$ that extends the involution $F \mapsto F^\perp$?

Answer. Yes. There is such an order-reversing involution. Further details will appear in [Wan26].

1 Context and motivation

Tropical linear subspaces of the tropical projective space $\mathbf{P}(\overline{\mathbb{R}}^n)$ correspond naturally and bijectively to valuated matroids on $[n]$, up to tropical rescaling, and a valuated matroid of rank k on $[n]$ has a valuated dual of rank $n - k$. The resulting involution is order-reversing on tropical linear subspaces of $\mathbf{P}(\overline{\mathbb{R}}^n)$.

One motivating example is the Boolean matroid $U_{n,n}$. Its lattice of flats $\mathcal{L}(U_{n,n})$ is the lattice of subsets of $[n]$, and $\text{Trop } U_{n,n}$ is the tropical projective space $\mathbf{P}(\overline{\mathbb{R}}^n)$. Duality for valuated matroids induces an involution on $\mathbf{Dr}(U_{n,n})$ that extends the natural involution $S \mapsto S^c$ on $\mathcal{L}(U_{n,n})$.

Another motivation comes from the classical geometry of linear subspaces and Grassmannians. Let V be a finite-dimensional vector space. A nondegenerate pairing on V induces an isomorphism $V \xrightarrow{\sim} V^\vee$, sending v to $v^\perp$. The map $W \mapsto W^\perp$ is an order-reversing involution on $\bigsqcup_k \mathbf{Gr}(k, V)$.

We note that matroids whose lattices of flats admit an order-reversing involution are called *modular matroids*. One does not expect any natural duality for tropical linear subspaces of tropical linear spaces that correspond to non-modular matroids.

2 Proof strategy

A valuated matroid μ of rank k on $[n]$ is a function $\binom{[n]}{k} \rightarrow \overline{\mathbb{R}}$ that satisfies the tropical Plücker relations:

- for each $S \subset [n]$ of size $k - 1$ and each $T \subset [n]$ of size $k + 1$,

$$\min_{a \in T \setminus S} \{ \mu(T - a) + \theta(S + a) \} \tag{2.1}$$

vanishes tropically, i.e., either all terms in (2.1) are infinity, or the minimum is achieved at least twice.

Here, $T - a := T \setminus \{a\}$ and $S + a := S \cup \{a\}$. We call (2.1) the *Plücker relation labeled by (S, T)* .

Tropical linear spaces correspond naturally and bijectively to valuated matroids, up to tropical scaling. For each valuated matroid θ of rank k , let $\text{Trop } \theta$ be the corresponding tropical linear space of dimension $k - 1$. A valuated matroid θ_1 is a quotient of θ_2 , denoted $\theta_2 \twoheadrightarrow \theta_1$, if they satisfy the tropical incidence-Plücker relations (3.3), and $\theta_2 \twoheadrightarrow \theta_1$ if and only if $\text{Trop } \theta_2 \supset \text{Trop } \theta_1$. We define

$$\mathbf{V}^k(\mu) := \{ \text{corank } k \text{ valuated matroid quotients of } \mu \}.$$

By construction, $\mathbf{V}^k(\mu) \subset \overline{\mathbb{R}}^{\binom{n}{r-k}}$, where r is the rank of μ .

Every matroid M gives rise to a valuated matroid, called the *trivial valuation on M* , that sends $B \in \binom{[n]}{r}$ to 0 if B is a basis of M , and to ∞ otherwise. We denote by $\mathbf{V}^k(M)$ the set of valuated matroid quotients of the trivial valuation on M and let $\mathbf{V}(M)$ be the disjoint union $\bigsqcup_k \mathbf{V}^k(M)$. Note that $\mathbf{V}(M)$ is partially ordered by the valuated matroid quotient relation. We construct a map $\theta \mapsto \theta'$ on $\mathbf{V}(M)$ that induces an order-reversing involution on $\mathbf{Dr}(M)$. The map is relatively simple, and can be viewed as the restriction of a tropical linear map in the natural Plücker coordinates.

In the analogous classical setting, the Plücker coordinates of $[W] \in \mathbf{Gr}(k, \mathbb{F}^r)$ are in bijection with coordinate subspaces of dimension k in $\mathbb{F}^r$. The matroid analog of a dimension k coordinate subspace is a rank k flat. Therefore, one might naturally guess that there should be a cryptomorphic characterization of valuated matroid quotients of rank k as functions on the rank k flats of M . We construct such a cryptomorphism in Proposition 3.4. The map Φ is then given simply by $\theta \mapsto \theta'$, where $\theta'(F) := \theta(F^\perp)$.

3 Solution

Throughout, M denotes a simple matroid on $[n]$.

3.1 A cryptomorphism of matroid quotients

Let N be a matroid on $[n]$ with rank function rk_N . Then N is a matroid quotient of M if and only if

$$\text{rk}_N(A + x) - \text{rk}_N(A) \leq \text{rk}_M(A + x) - \text{rk}_M(A) \quad (3.1)$$

for any $A \subset [n]$ and any $x \in [n]$. Lemma 3.1 is a cryptomorphic characterization of matroid quotients in terms of rank functions on $\mathcal{L}(M)$. The proof is straightforward and is omitted.

Lemma 3.1. *Let $\rho: \mathcal{L}(M) \rightarrow \mathbb{N}$ be any function that satisfies the following three properties*

- (Submodularity) $\rho(F) + \rho(G) \geq \rho(F \vee G) + \rho(F \cap G)$;
- (Unit increment) $\rho(F \vee x) - \rho(F) \in \{0, 1\}$;
- (Normalization) $\rho(\emptyset) = 0$.

Then $A \mapsto \rho(\text{cl}_M(A))$ is the rank function of a matroid quotient of M . Conversely, if ρ is the rank function of a matroid quotient of M , then the restriction of ρ to $\mathcal{L}(M)$ satisfies these three properties.

The proof is straightforward and omitted. We call the functions $\mathcal{L}(M) \rightarrow \mathbb{N}$ satisfying the properties in Lemma 3.1 *lattice rank functions*.

Let $\mathbf{Q}^k(M)$ be the set of corank k matroid quotients of M , with $\mathbf{Q}(M) := \bigsqcup_k \mathbf{Q}^k(M)$, partially ordered by the matroid quotient relation.

Proposition 3.2. *Let M be a modular matroid and $\rho: \mathcal{L}(M) \rightarrow \mathbb{N}$ be its lattice rank function. Then for any $N \in \mathbf{Q}^k(M)$ with lattice rank function rk_N , the function $\mathcal{L}(M) \rightarrow \mathbb{N}$ given by*

$$F \mapsto \rho(F) - \text{rk}_N([n]) + \text{rk}_N(F^\perp), \quad (3.2)$$

is the lattice rank function of some $N' \in \mathbf{Q}^{r-k}(M)$. The resulting map $N \mapsto N'$ is an order-reversing involution on $\mathbf{Q}(M)$, and $\Psi(M/F \oplus U_{0,F}) = M/F^\perp \oplus U_{0,F^\perp}$.

Proof. The proof that (3.2) is the lattice rank function of some corank k quotient $N' \in \mathbf{Q}^k(M)$ is identical to the proof of [JP18, Theorem 42]. A straightforward computation shows $(N')' = N$. Next, we show that $N'_2 \twoheadrightarrow N'_1$ whenever $N_1 \twoheadrightarrow N_2$. For any $F \subset G \in \mathcal{L}(M)$, we compute the difference

$$\text{rk}_{N'_2}(F) - \text{rk}_{N'_2}(G) - (\text{rk}_{N'_1}(F) - \text{rk}_{N'_1}(G)) = \text{rk}_{N_1}(F^\perp) - \text{rk}_{N_1}(G^\perp) - (\text{rk}_{N_2}(F^\perp) - \text{rk}_{N_2}(G^\perp)).$$

This is nonnegative because $N_1 \twoheadrightarrow N_2$, and hence $N'_2 \twoheadrightarrow N'_1$ by (3.1).

If rk is the lattice rank function of $M/F \oplus U_{0,F}$, then we have

$$\text{rk}(G) = \rho(G \vee F) - \rho(F).$$

Then (3.2) maps G to $\rho(G) + \text{rk}(G^\perp) - (n - \rho(F))$. Collecting terms, $G \mapsto \rho(G \vee F^\perp) - \rho(F^\perp)$, which is the lattice rank function of $M/F^\perp \oplus U_{0,F^\perp}$. $\square$

The following observation is immediate from the construction of Ψ .

Proposition 3.3. *The bases of N' are $\{B \mid \text{cl}_M(B)^\perp = \text{cl}_M(D) \text{ for some basis } D \text{ of } N\}$.*

3.2 A cryptomorphism of valuated matroid quotients

Let μ be a valuated matroid of rank r on $[n]$, whose underlying matroid is M . A valuated matroid θ of rank s on $[n]$ is a quotient of μ if and only if μ and θ satisfies the following tropical incidence-Plücker relations: for any $S \subset [n]$ of size $s - 1$ and $T \subset [n]$ of size $r + 1$,

$$\min_{a \in S \setminus T} \{\mu(T - a) + \theta(S + a)\} \quad (3.3)$$

vanishes tropically, i.e., either every term in (3.3) is infinity, or the minimum is achieved at least twice. We say (3.3) is the *incidence relation labeled by* (S, T) .

In what follows, we adopt a common convention from the literature on matroids and Grassmannians and use juxtapositions to denote unions, e.g. $SDab := S \cup D \cup \{a, b\}$. We will also need the following equivalent formulation of the tropical Plücker relations [BEZ21], called the *valuated basis exchange*. For any pair of bases (B_1, B_2) of μ , and for any $a \in B_1 \setminus B_2$, there is some $b \in B_2 \setminus B_1$ such that

$$\mu(B_1) + \mu(B_2) \geq \mu(B_1 - a + b) + \mu(B_2 - b + a).$$

Proposition 3.4. *Let μ be a valuated matroid on $[n]$ of rank r with underlying matroid M and θ be a valuated matroid on $[n]$ of rank $r - k$ with underlying matroid N . Then $\theta \in \mathbf{V}^k(\mu)$ if and only if it has the following properties.*

- (Degenerate flats) *If $\text{rk}_M(A) < r - k$, then $\theta(A) = \infty$.*
- (Parallel flats) *If $\text{cl}_M(A_1) = \text{cl}_M(A_2)$, and A_1D and A_2D are bases of M , then*

$$\mu(A_1D) - \mu(A_2D) = \theta(A_1) - \theta(A_2).$$

- (Concurrent flats) *Let $S \subset [n]$ be a subset of size $r - k - 1$ that is independent in M . For each circuit C of each contraction M/S , let D_C be a spanning set of M/S of size $k + 2$ containing C . Then*

$$\min_{a \in C} \{\theta(Sa) + \mu(SD_C - a)\} \quad (3.4)$$

vanishes tropically.

Proof. Suppose $\theta \in \mathbf{V}^k(\mu)$. We show θ satisfies the three properties in the proposition.

- (Degenerate flats) Let C_0 be a circuit in A . Let D_0 be a spanning set of M containing C_0 of size $r + 1$. Pick $x \in C_0$. The terms in the incidence relation labeled by $(A - x, D_0)$ are all infinity except for possibly $\theta(A) + \mu(D - x)$ so $\theta(A) = \infty$.
- (Parallel flats) Using basis exchange, it suffices to prove the case $|A_1 \setminus A_2| = 1$. Assume $A_1 = Sa$ and $A_2 = Sb$. Let D_1 be such that D_1Sa is a basis of M . Then the parallel flats property follows from the incidence relation (3.3) labeled by $(S, SDab)$.
- (Concurrent flats) Let $S \in \binom{[n]}{r-k-1}$ be independent. Let C be a circuit of the contraction M/S and D be a spanning set of M/S of size $k + 2$ containing C . Then (3.4) is simply the incidence relation labeled by (S, SD) .

Now, suppose θ is a valuated matroid that satisfies the three properties. Note that the parallel flats property and the concurrent flats property are exactly the nontrivial incidence relations labeled by (S, ST) for some S independent in M and some T .

Consider an arbitrary incidence relation for $\mu \rightarrow \theta$ labeled by (S, D) . We may assume S is independent in M , and that D does not contain a circuit of M that is contained in S . For contradiction, suppose the relation is not satisfied and the unique minimum is achieved uniquely by $\theta(S1) + \mu(D - 1) < \infty$. Since $S1$ is independent and $D - 1$ is a basis of M , there is some $T_1 \subset D - 1$ such that $S1T_1$ is a basis of M . Apply valuated basis exchange for μ to the pair of bases $(S1T_1, D - 1)$, we get some $c_1 \in D - 1$ such that

$$\mu(S1T_1) + \mu(D - 1) \geq \mu(ST_1c_1) + \mu(D - c_1).$$

All the values above are finite, so we can write

$$\mu(S1T_1) - \mu(ST_1c_1) \geq \mu(D - c_1) - \mu(D - 1).$$

Adding $\theta(Sc_1) - \theta(S1)$ (which, a priori, may be infinite) to both sides, we get

$$\theta(Sc_1) + \mu(S1T_1) - \theta(S1) - \mu(ST_1c_1) \geq \theta(Sc_1) + \mu(D - c_1) - \theta(S1) - \mu(D - 1) > 0,$$

so

$$\theta(Sc_1) + \mu(S1T_1) > \theta(S1) + \mu(ST_1c_1).$$

This implies, by the incidence relation labeled by $(S, S1T_1c_1)$, that there is some $b_1 \in T_1$ such that

$$\theta(Sc_1) + \mu(S1T_1) > \theta(S1) + \mu(ST_1c_1) \geq \theta(Sb_1) + \mu(S1T_1 + c_1 - b_1).$$

Set $T_2 = T_1 + c_1 - b_1$. Since $\theta(Sb_1) + \mu(ST_1c_1) < \infty$, $\theta(Sb_1) + \mu(S1T_2) < \infty$. Suppose we have obtained T_ℓ this way. Since $\mu(S1T_\ell) < \infty$, we can apply valuated basis exchange to the pair $(S1T_\ell, D - 1)$, we get some $c_\ell \in D - 1$ such that

$$\mu(S1T_\ell) + \mu(D - 1) \geq \mu(ST_\ell c_\ell) + \mu(D - c_\ell),$$

where all the terms are finite. Again, this implies

$$\theta(Sc_\ell) + \mu(S1T_\ell) > \theta(S1) + \mu(ST_\ell c_\ell)$$

and by the incidence relation labeled by $(S, S1T_\ell c_\ell)$, there is some $b_\ell \in T_\ell$ such that

$$\theta(Sc_\ell) + \mu(S1T_\ell) > \theta(S1) + \mu(ST_\ell c_\ell) \geq \theta(Sb_\ell) + \mu(S1T_\ell + c_\ell - b_\ell).$$

Now set $T_{\ell+1} = T_\ell + c_\ell - b_\ell$. For each ℓ , we have inequality

$$\theta(Sc_\ell) + \mu(S1T_\ell) > \theta(Sb_\ell) + \mu(S1T_{\ell+1}),$$

where the right hand side is finite. Since $T_\ell \subset D$ for any ℓ , there are only finitely many possibilities for T_ℓ , so for some $\ell_2 > \ell_1$, we have $T_{\ell_1} = T_{\ell_2}$. We have inequality of the sum

$$\sum_{i=\ell_1}^{\ell_2-1} \left(\theta(Sc_i) + \mu(S1T_i) \right) > \sum_{i=\ell_1}^{\ell_2-1} \left(\theta(Sb_i) + \mu(S1T_{i+1}) \right). \quad (3.5)$$

Since $T_{i+1}b_i = T_i c_i$, we have equality of multi-sets

$$\bigsqcup_{i=\ell_1}^{\ell_2-1} (T_{i+1}b_i) = \bigsqcup_{i=\ell_1}^{\ell_2-1} (T_i c_i).$$

Thus $\{b_{\ell_1}, \dots, b_{\ell_2-1}\} = \{c_{\ell_1}, \dots, c_{\ell_2-1}\}$ as multi-sets, and the two sides of (3.5) are equal, a contradiction. $\square$

3.3 Construction of Φ

For each $\theta \in \mathbf{V}^k(M)$, we define a function $\Phi(\theta): \mathcal{L}^{r-k}(M) \rightarrow \overline{\mathbb{R}}$ by

$$I \mapsto \begin{cases} \theta(J), & \text{if } \text{cl}_M(J) = \text{cl}_M(I)^\perp, \\ \infty, & \text{otherwise.} \end{cases}$$

It is clear that Φ is compatible with $F \mapsto F^\perp$. We now show that $\Phi(\theta)$ is a rank k valuated quotient of M .

The tropical linear space $\text{Trop } M$ is the intersection of the tropical hyperplanes corresponding to the circuits of M . A *tropical basis* of M is a subset of the circuits whose corresponding tropical hyperplanes cut out $\text{Trop } M$. Recall the *pasting* operation from [YY07]: if C_1 and C_2 are two circuits of M with $C_1 \cap C_2 = \{a\}$ then pasting them means taking their symmetric difference $(C_1 \cup C_2) \setminus \{a\}$. If every circuit in M can be constructed by successive pasting of circuits in a collection $\mathcal{S}$, then $\mathcal{S}$ is a tropical basis [YY07, Lemma 4].

Proposition 3.5. *For any simple modular matroid, the 3-element circuits form a tropical basis.*

Proof. Every simple modular matroid is a direct sum of a free matroid and finite projective geometries. We only need to prove the statement for M a connected finite projective geometry. If $M = U_{2,n}$, all circuits have size 3. Assume M is a finite projective geometry of rank ≥ 3 . We will show that the 3-element circuits of M generate all circuits via pasting. We proceed by induction on the rank of M .

If M has rank 3, then a size 4 circuit $\{a_1, a_2, b_1, b_2\}$ is a symmetric difference between two circuits $\{a_1, a_2, c\}$ and $\{b_1, b_2, c\}$, where c is the point where the line $\text{cl}_M(a_1 a_2)$ and the line $\text{cl}_M(b_1 b_2)$ intersect. Let M have rank $r > 3$. By the induction hypothesis, any circuit of size $\leq r$, which is a circuit of a finite projective geometry of lower rank, can be obtained by pasting the 3-element circuits. Let C be a circuit of size $r + 1$. Take two distinct points $a, b \in C$ and let $C' = C \setminus \{a, b\}$. The line $\text{cl}_M(ab)$ intersects with $\text{cl}_M(C')$ at a point d . Note that d is not in the closure of any proper subsets of C' , otherwise C could not be a circuit. This means both $C' \cup \{d\}$ and $\{a, b, d\}$ are circuits. Since C is the symmetric difference of C' and $\{a, b, d\}$, the claim follows by induction. $\square$

Proposition 3.6. *If μ is the trivial valuation on a modular matroid M , then the concurrent flats property in Proposition 3.4 can be replaced by the following condition,*

- (Concurrent triples) *Let $S \in \binom{[n]}{r-k-1}$ be independent in M . For each circuit $C = \{a, b, c\}$ of M/S ,*

$$\min\{ \theta(Sa), \quad \theta(Sb), \quad \theta(Sc) \} \tag{3.6}$$
vanishes tropically.

In other words, a valuated matroid θ on $[n]$ of rank $r - k$ is in $\mathbf{V}(M)$ if and only if it satisfies the degenerate flats property, the parallel flats property, and the concurrent triples property.

Proof. The contraction M/S is a modular matroid, so the statement follows from Proposition 3.5. $\square$

We say that three distinct flats $F_1, F_2, F_3 \in \mathcal{L}(M)$ form a *concurrent triple* if they cover the same flat G . The following corollary follows easily from the parallel flats property and the concurrent triples property.

Corollary 3.7. *Let A_1, A_2 and A_3 be independent sets of M such that $\text{cl}_M(A_1), \text{cl}_M(A_2)$ and $\text{cl}_M(A_3)$ form a concurrent triple of corank k flats. Then for any $\theta \in \mathbf{V}^k(M)$,*

$$\min\{ \theta(A_1), \quad \theta(A_2), \quad \theta(A_3) \} \tag{3.7}$$

vanishes tropically.

Let N be the underlying matroid of $\theta \in \mathbf{V}^k(M)$. Denote $\Phi(\theta)$ by θ' . Note that the support of θ' is the set of bases of N' , as in Proposition 3.2.

The degenerate flats properties and the parallel flats properties for θ' are clear. To show that θ' is in $\mathbf{V}^{r-k}(M)$, it is enough to verify that it satisfies concurrent triples properties, by Proposition 3.6. Suppose $A \in \binom{[n]}{k-1}$ is independent in M and $C = \{1, 2, 3\}$ is a 3-element circuit of the contraction M/A . Let

$F := \text{cl}_M(A)$, $F_i := \text{cl}_M(Ai)$ for each $i \in \{1, 2, 3\}$, and $F_{123} = \text{cl}_M(A123)$. Then F_1, F_2 and F_3 form a concurrent triple. By the modularity of M , $F_1^\perp, F_2^\perp$, and $F_3^\perp$ form a concurrent triple. It then follows from Corollary 3.7 that

$$\min\{\theta'(A_1), \theta'(A_2), \theta'(A_3)\}$$

vanishes tropically.

Now, we show that θ' is a valuated matroid. Since the support of θ' is a matroid, it suffices to verify that θ' satisfies all the 3-term Plücker relations [Mur10, Theorem 5.2.25]. Choose $S \in \binom{[n]}{k-2}$ and, by relabeling, assume $1, 2, 3, 4 \notin S$. We need to show

$$\min\{\theta'(S12) + \theta'(S34), \theta'(S13) + \theta'(S24), \theta'(S14) + \theta'(S23)\} \quad (3.8)$$

vanishes tropically. Define

$$H := \text{cl}_M(S), \quad H_a := \text{cl}_M(Sa), \quad H_{ab} := \text{cl}_M(Sab), \quad H_{1234} := \text{cl}_M(S1234), \quad (3.9)$$

where $a, b, c, d \in \{1, 2, 3, 4\}$ are distinct elements. Up to relabeling, we may assume that $S12$ and $S34$ are both independent. We carried out the cases according to the contraction $M' = M|_{S1234}/S$. Let $\mathcal{P}$ be the set of all the flats in (3.9).

- Case 2: $M' \cong U_{3,4}$. Then any $\{b, c, d\} \subset \{1, 2, 3, 4\}$ is a circuit of $M|_{Sabcd}/Sa$. We have just shown that

$$\min\{\theta'(Sab), \theta'(Sac), \theta'(Sad)\}$$

vanishes tropically by the concurrent triple property for any distinct $a, b, c, d \in \{1, 2, 3, 4\}$. Now it follows from [JLLO23, Theorem 3.10] that (3.8) vanishes tropically.

- Case 3: $M' \cong U_{2,2} \oplus U_{1,2}$. We may assume $H_{12} = H_1 = H_2$. Then $\theta'(S12) = \infty$ and (3.8) becomes

$$\min\{\theta'(S13) + \theta'(S24), \theta'(S14) + \theta'(S23)\},$$

which vanishes tropically, because $\theta'(S13) = \theta'(S23)$ and $\theta'(S14) = \theta'(S24)$.

- Case 4: $M' \cong U_{2,3} \oplus U_{1,1}$. We may assume $H_{12} = H_{13} = H_{23}$. Then $\theta'(S12) = \theta'(S13) = \theta'(S23)$. Note that (3.8) follows, because the concurrent triples property says

$$\min\{\theta'(S14), \theta'(S24), \theta'(S14)\}$$

vanishes tropically.

- Case 5: M' has rank 2. Recall that we assume $S12$ and $S34$ are both independent. By the basis exchange property, we may also assume, by relabeling, that $S13$ and $S24$ are both independent. Then we have $H_{12} = H_{34} = H_{13} = H_{24}$. Either $\text{cl}_M(S14) = H_{12}$ or $S14$ is dependent. Hence, $\theta'(S14)$ is either equal to $\theta'(S12)$ or infinity, meaning (3.8) vanishes tropically.

It remains to show that Ψ reverses the partial order on $\mathbf{V}(M)$.

Proposition 3.8. *For any $\theta_1 \in \mathbf{V}_t(M)$ and $\theta_2 \in \mathbf{V}_s(M)$ where $t \geq s$, $\theta_1 \rightarrow \theta_2$ if and only if $\Phi(\theta_2) \rightarrow \Phi(\theta_1)$.*

Proof. Denote $\Phi(\theta)$ by θ' . Since Φ is an involution on $\mathbf{V}(M)$, it suffices to show that $\theta'_2 \rightarrow \theta'_1$ implies $\theta_1 \rightarrow \theta_2$. By Proposition 3.2 and Proposition 3.3, the underlying matroid of θ_2 is a quotient of the underlying matroid of θ_1 . This means the degenerate flats property is satisfied. By Proposition 3.4, it remains to show that the parallel flats property and the concurrent flats property are satisfied. Recall that those are exactly the incidence relations for $\theta_1 \rightarrow \theta_2$ labeled by (S, ST) where $|S| = s - 1$ and $|T| = t - s + 2$. By Proposition 3.4, we may assume $\text{rk}_M(ST) \geq t$ for the relation to be nontrivial.

- Case 1: $\text{rk}_M(ST) = t$. There is a unique circuit C of $M|_{ST}/S$ and T is a spanning set of $M|_{ST}/S$ that contains C . Since $\text{cl}_M(ST - a) = \text{cl}_M(ST)$ for any $a \in C$, by the parallel flats property, $\theta_1(ST - a)$ does not depend on $a \in C$. Hence

$$\min_{a \in C}\{\theta_1(ST - a) + \theta_2(S + a)\}$$

vanishes tropically, by the concurrent flats property satisfied by θ_2 that $\min_{a \in C}\{\theta_2(S + a)\}$ vanishes tropically.

- Case 2: $\text{rk}_M(ST) = t + 1$. For each $I \subset T$, let $F_I = \text{cl}_M(SI)$. Notice that ST is a basis of $\text{rk}_M(ST)$, so $F_I \cap F_J = F_{I \cap J}$ and one has a Boolean sublattice

$$\mathcal{P} := \{F_I \mid I \subset T\}.$$

The dual of this sublattice

$$\mathcal{P}^\perp := \{F_I^\perp \mid I \subset T\}$$

is also a Boolean sublattice. Let S' be a basis of $F_\emptyset^\perp$. Then there is some T' of size $t - s + 2$ such that $\mathcal{P}^\perp$ is the Boolean sublattice

$$\{G_I := \text{cl}_M(S'I) \mid I \subset T'\}.$$

We thus obtain a bijection $\sigma: T \rightarrow T'$ by setting $F_{\{a\}} \mapsto F_{\{a\}}^\perp = G_{T' \setminus \{\sigma(a)\}}$. By definition, $\theta_1(ST - a) = \theta'_1(S' + \sigma(a))$ and $\theta_2(S + a) = \theta'_1(S'T' - \sigma(a))$. Hence, the incidence relation for $\theta_1 \rightarrow \theta_2$ labeled by (S, ST) is the same as the incidence relation labeled for $\theta'_2 \rightarrow \theta'_1$ by $(S', S'T')$, and is thus satisfied.

Since in either case, the incidence relations for $\theta_1 \rightarrow \theta_2$ are satisfied, the statement is proved. $\square$

We have shown that Φ maps $\mathbf{V}^k(M)$ to $\mathbf{V}^{r-k}(M)$ and is order-reversing. It is then straightforward to check that Φ is also an involution, which completes the solution.

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